2.10.52

EE25BTECH11023 - Venkata Sai

Question:
Let $\mathbf{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{c} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$. A vector in the plane of $\mathbf{a}$ and $\mathbf{b}$ whose projection on $\mathbf{c}$ is $\frac{1}{\sqrt{3}}$, is

1) $4\hat{\mathbf{i}} - \hat{\mathbf{j}} + 4\hat{\mathbf{k}}$
2) $3\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}}$
3) $2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$
4) $4\hat{\mathbf{i}} + \hat{\mathbf{j}} - 4\hat{\mathbf{k}}$

Solution:
Given

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \tag{1}$$

Let $\mathbf{r}$ be coplanar to $\mathbf{a}$ and $\mathbf{b}$

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \tag{2}$$

Given the projection of $\mathbf{r}$ on $\mathbf{c}$ is $\frac{1}{\sqrt{3}}$

$$\frac{\mathbf{r} \cdot \mathbf{c}}{\|\mathbf{c}\|} = \frac{1}{\sqrt{3}} \tag{3}$$

$$\mathbf{r} \cdot \mathbf{c} = (\mathbf{a} + t\mathbf{b}) \cdot \mathbf{c} \tag{4}$$
$$= \mathbf{a} \cdot \mathbf{c} + t\,(\mathbf{b} \cdot \mathbf{c}) \tag{5}$$
$$= \mathbf{a}^\top \mathbf{c} + t\left(\mathbf{b}^\top \mathbf{c}\right) \tag{6}$$

On substituting the given values,

$$\mathbf{r} \cdot \mathbf{c} = \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + t\left(\begin{pmatrix} 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}\right) \tag{7}$$
$$= 1 + 2 - 1 + t\,(1 - 1 - 1) \tag{8}$$
$$= 2 - t \tag{9}$$

$$\|\mathbf{c}\|^2 = \mathbf{c}^\top \mathbf{c} = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \tag{10}$$
$$= 1 + 1 + 1 = 3 \tag{11}$$

$$\frac{|2 - t|}{\sqrt{3}} = \frac{1}{\sqrt{3}} \tag{12}$$

yielding

$$(2 - t) = \pm 1 \tag{13}$$

$$\implies t = 1 \text{ or } 3 \tag{14}$$

When t=1, we have

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \tag{15}$$

When t=3, we have

$$\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 4 \end{pmatrix} \tag{16}$$

Hence Option(1) is the correct answer

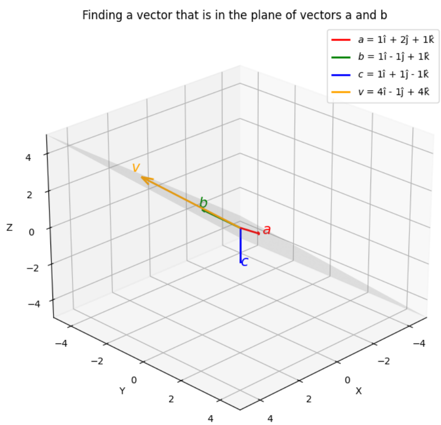

Fig. 4.1